

SEDAR 10: Gulf of Mexico and South Atlantic Gag Grouper Gulf of Mexico Gag Grouper

Data Workshop

- Life History
 - * Conduct further review of current sampling methodologies by sector, including detailed comparison of length data from otolith samples and from more expansive port-based length sampling (via TIP; see SEDAR10-DW24).
 - * Bring increased attention to the need for strategies to improve port sampling (representation of fishery sectors and random sampling)
 - * Increase the sampling of the recreational sector for biological samples throughout the docks and ports of Florida's west coast.
 - * Continue support of fishery-independent surveys including all gears (hand-line, long-line, and trap) throughout the west Florida shelf.
 - * Recognize that gag landings may be increasing elsewhere in the Gulf and bring increased attention to sampling the northern and western Gulf regions.
 - * Continue exchanges of calibration otoliths sets and age workshops among state and federal agencies, and universities to continue improvements of data comparability and quality control.
 - * The DW recommends continued research on the use of otolith chemistry to evaluate the population structure of gag.
 - * Continue genetics research to determine connectivity among different regions. The DW further highly recommends every opportunity be taken to add Mexican (Campeche) samples to this analysis as these methods can be most informative in divining patterns of gene flow and population connectivity.
 - * The DW suggests that it may be particularly valuable to convene a workshop to address the potential non-random and non-representative sampling that hampers collection of small numbers of biological samples (relative to numbers of fish landed) which in turn are used for parameter estimates.
 - * The DW recommends that age structure sampling continue on an annual basis in the Gulf.

- * The DW recommends that larval transport and modeling efforts associated with development of an Integrated Coastal Ocean Observing System (ICOOS) is further supported.
- * Tagging studies are needed to: 1) clarify the extent of movement between the Gulf and SA regions and within region, and 2) aid further development of age-specific estimates of depth-related mortality in the Gulf region. In the Gulf region, we recommend that tagging effort be extended to the middle and outer shelf, perhaps with the assistance of cooperating commercial fishers, for the purpose of tagging adult gag. The DW recommends that future tagging studies should be done in a more coordinated manner between researchers in the Gulf and SA regions, particularly with respect to gear, fish size, and depth.
- Commercial Statistics
 - * Increase sampling for otoliths for aging
 - * Improve at-sea observation for discards
- Recreational Statistics
 - * Recommended a closer examination of reported headboat fishing locations, with respect to the GMFMC-SAFMC dividing line.
 - * Explore surrogates for recreational fishing effort, for example numbers of recreational boat licenses or numbers of operating headboats.
 - * MRFSS shore mode be explored further to elucidate whether it provides a useful annual signal of catches.
- Indices of Abundance
 - * Develop a suitable method to correct species misidentification between black and gag grouper on a trip by trip basis.
 - * The group strongly recommends increased adequate funding for both developing new and maintaining existing fishery-independent sampling programs, and stresses that quality indices require continuous funding over meaningful time periods (ideally decades).
 - * When possible, environmental factors should be considered in future index standardization procedures.
 - * The group recognized the need to quantify changes in catchability over time.

- * Recommend the use of an assessment model structure that can accommodate a nonlinear relationship between CPUE indices and stock size. Since data are often lacking, the group recommends sensitivity analyses that fix the nonlinear parameter(s) at plausible values.

Assessment Workshop

No research recommendations provided.

Review Workshop

- Age determination: The Review Panel noted the importance of age reading comparisons and recommended that exchange of otoliths between labs continue in the future.
- Stock structure: The Review Panel recommended a further examination of stock structure before the next assessment, including a detailed analysis of existing tagging data and the initiation of new tagging experiments.
- The Panel recommends that a special workshop be convened to estimate and quantify changes in catchability over the last 25 to 30 years.

South Atlantic Gag Grouper

Data Workshop

- Life History
 - * Continue annual sampling for age structure with increased attention to representative sampling.
 - * Continue exchanges of calibration otoliths sets and age workshops among state and federal agencies, and universities to continue improvements of data comparability and quality control.
 - * The DW recommends continued research on the use of otolith chemistry to evaluate the population structure of gag.
 - * Continue genetics research to determine connectivity among different regions. The DW further highly recommends every opportunity be taken to add Mexican (Campeche) samples to this analysis as these methods can be most informative in divining patterns of gene flow and population connectivity.
 - * The DW suggests that it may be particularly valuable to convene a workshop to address the potential non-random and non-representative sampling that hampers collection of small numbers of biological samples (relative to numbers of fish landed) which in turn are used for parameter estimates.

- * The DW recommends that long-term continuous monitoring of age structure be undertaken in the South Atlantic to test this hypothesis that strong year classes are reflected in both the South Atlantic and Gulf of Mexico.
- * The DW recommends that larval transport and modeling efforts associated with development of an Integrated Coastal Ocean Observing System (ICOOS) is further supported.
- * Tagging studies are needed to: 1) clarify the extent of movement between the Gulf and SA regions and within region, and 2) aid further development of age-specific estimates of depth-related mortality in the Gulf region. In the SA region, most of the tagging effort has been off South Carolina. Therefore, we recommend that additional tagging be completed off the east coast of Florida to examine the extent of northerly and southerly movements. The DW recommends that future tagging studies should be done in a more coordinated manner between researchers in the Gulf and SA regions, particularly with respect to gear, fish size, and depth.
- Commercial Statistics
 - * Increase sampling for otoliths for aging
 - * Improve at-sea observation for discards
 - * Continued education of samplers for species identification
 - * Conversions needed for different market categories (gutted, headed, filleted, whole weight).
- Recreational Statistics

No research or monitoring recommendations provided.
- Indices of Abundance
 - * Investigate further the issue of misidentification between black grouper and gag. Develop a suitable method to correct misidentifications on a trip by trip basis. This issue will also be of concern when assessing black grouper. The catches of gag grouper misidentified as black is likely a substantial proportion of reported black grouper landings.
 - * We recognize that many valuable and well designed fishery-independent sampling programs have been underfunded or discontinuously funded, resulting in low sample sizes, variable sampling effort (in time and space), discontinuous time series, and poorly stratified designs. The group strongly

recommends increased funding toward developing and maintaining fishery-independent sampling programs, and stresses that quality indices require continuous funding over meaningful time periods (ideally decades).

- * It was proposed that the index working group examine the possibility of including environmental variables in computation of indices. Variables discussed included wave height, sea surface temperature, surface currents and hurricane impact. The group considered that other model parameters, particularly the spawner-recruit relationship, might be a meaningful way to include environment variables in assessment models.
- * Examine methods to account for changes in catchability over time of abundance. This is of particular importance when considering fisheries-dependent indices.
- * Develop coast-wide sampling of larval and juvenile abundance.

Assessment Workshop

- The AW recommends that spatial information, including the depth related mortality functions suggested by the DW, continue to receive research attention.
- Improved spatial information on gag grouper to be used for depth related mortality functions (DW suggestion that could not be implemented for the south Atlantic assessment), and to monitor for potential changes in range that may affect assessment results.
- The AW also recommends that data be collected in the South Atlantic on effort and discards by depth.
- The AW recommends a fishery independent index of abundance be developed. A major missing component is the availability of a fishery independent index, as all three available indices were fishery dependent and therefore subject to shifts in efficiency and regulations.
- The AW recommends that the gag grouper mature sex ratio needs to be observed, from which it may also be possible to infer information about male fertility and the number of sperm required for successful fertilization. The potential results of shifts in sex ratio in a protogynous species like gag are not entirely known.
- The AW recommends further examination and reconstruction of the catch and total removals history (prior to 1962) from data sources not currently contributing the assessment history.

- The AW suggests that methods like DNA tagging may prove useful as a means for gaining an independent snapshot of total mortality. Estimates of mortality may be difficult to attain or determine if current estimates are on the correct scale.
- The AW recommends that effectiveness of effort from technological changes (e.g., electronics, GPS) be examined. The assessment ran alternate base runs that both assumed increasing catchability from improvements in technology and no increases in catchability. The AW agreed that this increase in technology had occurred, though any level had to be heavily inferred from studies in other fisheries. Research should be conducted in the major grouper fisheries to determine a more appropriate level and degree of increasing catchability.

Review Workshop

- The Panel recommends that a special workshop be convened to estimate and quantify changes in catchability over the last 25 to 30 years.
- Strengthen the MRFSS program to provide more precise estimations of the age/length composition.
- Provide more detailed model diagnostics, such as complete lists of estimated parameters together with their estimated standard errors, in model sensitivity runs.
- Explore the model residuals diagnostics to test for time series autocorrelation contributions to the lack of goodness of fit in the assessment.
- Analyze the existing mark-recapture data and initiate new mark-recapture studies, which will help identify movements and migrations between two stocks, estimate fishing mortality, enhance population estimates; and better identify the stock structure and habitat preferences.
- Bias on estimating weight from the log-log length-weight relationship

General Assessment Advice From the Assessment Workshop

- Never rely on any one assessment procedure.
- Include retrospective analyses showing how estimates change with time.
- Beware of complex size-age and temporally changing vulnerability schedules.
- Beware of confounding between stock-recruitment and recruitment anomaly (environmental) effects.

- Examine implications of relative abundance time series that give contradictory indications of time trends.
- Provide time series estimates of fishing mortality rates.
- Run assessments on the longest possible catch data series, to give the best possible long term perspective on stock status.
- Carefully examine any available spatial data for evidence of range collapse or expansion.

Review Workshop Recommendations for both Stocks

- There was large volume of documentation associated with this RW. The Review Panel recommends a clear executive summary for all substantive Data and Assessment Documents.
- It could be more informative to distribute a succinct table of model equations and parameters (estimated and observed) to be provided for each assessment along with, if appropriate, a table of management options (e.g. a decision table) and the risks associated with them.

CIE Contractor Recommendations

Research and Assessment Recommendations

- Information on the number, location and persistence of spawning aggregations should be obtained and presented in future assessments in order to identify essential habitat (if this information is not already available).
- A further examination of stock structure should be completed before the next assessment, including a detailed analysis of existing tagging data and, possibly, the initiation of new tagging experiments to estimate mixing rates and the associated fishing mortality independent of the commercial fishing. This would necessitate an effective design for estimating tagging mortality, tagging shedding, reporting rates to increase confidence in the stock assessments.
- Standard fisheries methods based on yield per recruit analyses may not be appropriate for species that change gender during their lifetime. Spawner recruit analyses should consider males and females reproductive biomasses separately. In the case of gag grouper, male biomass may become limiting before female biomass does. In this context, projections of future population status should be provided by gender in the next assessment.

SEDAR Process Recommendations

- Like the SAW process, the SEDAR Review Workshop is now reliant solely upon panelists provided by the Center for Independent Experts. In my opinion, this poses some concerns. Under the former model (e.g.: SEDAR6), the Review Panel consisted of scientific experts from the CIE, from the NMFS, and from academia. This provided for a broader expertise in the review process. The current model is designed to assess scientific credibility only and not to provide management advice. This is a positive step as it provides a buffer between the science of stock assessment and the potential politics of management. This buffer or barrier should be maintained and the revised model attempts to address this. However, the assessment of scientific credibility should not preclude additional panellists besides those provided by the CIE.
- The assessment of each of the stocks was conducted by separate teams, using similar but somewhat different assessment models. It was therefore more difficult for the Review Panel to make direct comparisons between assessment results. Recognizing that this was the first time that either of these stocks was assessed under the SEDAR process, the assessment teams did an excellent job. However, in future, a more thorough review could be facilitated if the assessment teams worked cooperatively using a single model for both stocks.